Stage 1 Document: Requirements Specification and Project Scope

Engineering Project Documentation
Faculty of Mathematics and Information Science
Warsaw University of Technology

---

1. Document Metadata and Change History

Version	Date	Author	Description of Change
1.0	15.10.2024	Alice Kowalska	Baseline version for supervisor sign-off
1.1	18.10.2024	Bob Nowak	Added quantifiable non-functional requirements (NFRs)

- **Project Title:** *A System for Reconstructing Fragmented Documents Based on the Shape and Content of Their Parts*
- **Authors:** Martyna Sadowska, Hanna Szczerbińska, Zuzanna Zalewska
- **Supervisor:** Janusz Rafałko, Ph.D.

---

2. One-Sentence Engineering / Research Problem Formulation

> *"The objective of this project is to design, implement, and evaluate a software system capable of reconstructing fragmented (shredded or torn) documents based on an image of their pieces, supported by a dedicated document-splitting module for synthetic dataset generation and testing."*

---

3. Project Scope: MVP vs. Future Work (Scope Management)

3.1. Minimum Viable Product (MVP – Passing Scope)

1. **Document Splitting Module (Module 1):**
   - Supports single and batch mode processing.
   - Allows users to split document images using various algorithms, preview piece borders, re-generate if needed, and apply random rotations and non-overlapping background placement.
   - Saves the generated fragments locally.

2. **Document Reconstruction Module (Module 2):**
    - Extracts individual fragments from an input image (single and batch mode).
    - Reconstructs original documents using selectable or automated algorithms.
    - Displays the reconstructed output, allows iterative refinement (adjusting time/algorithm), and saves the final result locally.

3.2. Out of Scope (Future Work)

1. Integration with external cloud storage providers or dedicated mobile applications.

---

4. Functional Requirements (User Stories & Acceptance Criteria)

US-01: Ingestion of a Single Telemetry Data Point

- **As an:** External system (e.g., IoT sensor / telemetry script),
- **I want to:** Submit a measurement with a timestamp and numeric value via the `POST /api/v1/metrics` endpoint,
- **So that:** The data is validated, stored in the time-series database, and forwarded to the analytical engine.

  Acceptance Criteria (Gherkin):

  - **Given:** A running and authenticated ingestion service,
  - **When:** The client sends a `POST` request with a valid JSON payload `{ "sensor_id": "temp-01", "timestamp": 1729000000, "value": 24.5 }`,
  - **Then:** The system returns response status `201 Created` in ≤ 50 ms, and the record is persisted in the TimescaleDB hypertable.
  - **Scenario (Invalid data):**
  - **When:** The client sends a request with a negative timestamp or a missing `value` field,
  - **Then:** The system rejects the request with HTTP `422 Unprocessable Entity` containing a detailed validation error message.

US-02: Viewing Time-Series Chart with Flagged Anomalies

- **As an:** Operations engineer / data analyst,
- **I want to:** Open the chart view for a selected sensor covering the last 24 hours with highlighted anomalies,
- **So that:** I can quickly diagnose equipment faults across infrastructure.

Acceptance Criteria (Gherkin):

- **Given:** A user logged in to the web dashboard,
- **When:** The user selects sensor `sensor_id="temp-01"` and time window `24h` from the menu,
- **Then:** The line chart renders in < 1.0 s with anomalous points highlighted in red.

---

5. Quantifiable Non-Functional Requirements (NFRs)

ID	Category	Target Quantifiable Metric	Verification Method
NFR-01	**Response Time (Latency)**	95% of write and read requests respond in$<$ 200 ms under a load of 50 concurrent virtual users.	k6 / Locust load benchmark
NFR-02	**Throughput**	Ingestion engine processes$\geq$ 5 000 events/second on a 4-core machine with 16 GB RAM.	Locust stress test
NFR-03	**Resource Utilization**	Total idle RAM consumption of the container stack$\leq$ 2.0 GB; analytical model runs in CPU memory without GPU requirements.	`docker stats`
NFR-04	**Deployment Reproducibility**	Complete deployment and startup of a clean environment using`docker compose up --build` completes in $<$ 3 minutes on a MiNI lab Ubuntu machine.	Stopwatch / installation script
NFR-05	**Reliability**	0 unhandled HTTP 500 exceptions on malformed input data; automated database reconnection retries in containers.	pytest integration test suite

---

6. Glossary

Term	Definition
APD (Archiwum Prac Dyplomowych)	University Thesis Archive; official web portal of Warsaw University of Technology used for thesis submission, anti-plagiarism verification, and archiving.

Term	Definition
Hypertable	A TimescaleDB abstraction that presents time-series data as a single continuous PostgreSQL table while transparently partitioning it into time-based chunks under the hood.
Isolation Forest	An unsupervised machine learning algorithm for anomaly detection that isolates observations by randomly selecting a feature and split value, where anomalies require fewer splits to isolate.
MVP (Minimum Viable Product)	The minimal functional scope of the project fulfilling all foundational requirements for academic project evaluation and defense.
p95 Latency	The 95th percentile of response time distribution, indicating that 95% of all incoming requests are served in less than the specified duration.
Rolling Z-Score	A localized statistical metric representing the number of standard deviations a given observation deviates from the moving mean over a sliding temporal window.
Telemetry	Automated remote collection and transmission of operational measurement data from distributed sensors or agents to a central receiving station.

---

7. Preliminary Risk Register and Role Allocation

(Detailed Risk Register and RACI Matrix are defined in dedicated documents: `management/risk-register-and-contingency.md` *and* `management/team-charter-and-raci.md`*.)*

---

8. Formal Supervisor Acceptance

☒ **Scope and MVP Sign-off:** The supervisor has reviewed the requirements specification and approves the MVP scope as the foundation for the engineering / master's degree project.

Supervisor Signature: .....
Date: October 18, 2026